

TITLE PLACEHOLDER

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Abstract

ABSTRACT PLACEHOLDER

Introduction

Suppose that you attend a friend's wedding. That evening you describe it to your partner, recounting how the ceremony ran long, how the best man's toast was unexpectedly moving, and how a small child wandered onto the dance floor midway through the first song. A week later a coworker asks how the wedding was, and you tell the story again. What is that second telling a telling of? One appealing answer is that both tellings draw on the same underlying memory of the wedding, and that whatever differences arise between them reflect the ordinary imprecision of putting an experience into words. A second possibility, which motivates the work we report here, is that your first telling has itself become part of what you are remembering.

What happens to an experience between living it and describing it later? Since our brains do not maintain an exhaustive account of everything we encounter, the very act of perceiving an experience already necessarily entails compressing and distorting it. We attend closely to some

17 aspects of a scene while ignoring others, and we tend to interpret ambiguous moments in light
18 of what has happened previously and we expect will happen next. But once an experience has
19 ended, neither the experience itself nor our moment-to-moment perception of it remains available,
20 to us or to anyone else. What remains is instead the product of a further distillation, in which
21 our perception of the experience is reduced to something our memory systems can store and
22 later reconstruct. Which aspects of an experience survive this second distillation may depend on
23 what we attended to, what we anticipated would matter later, what we found interesting, and the
24 thoughts, motivations, and biases we brought with us. Because remembering entails reconstructing
25 an experience rather than replaying it, the resulting account often preserves the general “shape”
26 of what happened while discarding many of its particulars (Bartlett, 1932; Schacter, 2012; Schacter
27 et al., 2011). The units over which these distillations operate appear to be events, in that people
28 segment continuous experience into discrete episodes at moments of change, and those boundaries
29 shape what is remembered afterwards (Baldassano et al., 2017; Radvansky and Zacks, 2017). In
30 prior work we characterized this transformation geometrically by modeling an experience and a
31 person’s recounting of it as trajectories through a semantic space, and asking which properties of
32 the experience’s trajectory are preserved in the recounting (Heusser et al., 2021).

33 Yet memories do not remain fixed once they have been encoded, pointing to further transfor-
34 mations that a single recounting cannot reveal. Information may be lost over time; experiences we
35 have after an event may color how we later remember it; similar experiences may interfere with
36 one another; and we may simply re-evaluate what happened in light of what we have learned
37 since. The act of retrieving a memory itself appears to be one of the forces that changes it, in that
38 a memory can become labile when it is reactivated and must be stabilized again afterwards (Hup-
39 bach et al., 2007, 2009; Nader et al., 2000; Sederberg et al., 2011). Retrieval also changes what will
40 be accessible later—often substantially—relative to memories that go unretrieved (Karpicke and
41 Roediger, 2007; Roediger and Karpicke, 2006). Further, the particular way in which we recount
42 an experience appears to matter as well, since people who are asked to describe an event for one
43 purpose later remember it in ways that are biased toward that description (Tversky and Marsh,
44 2000). Memories also become more semantic and less perceptually detailed, both with the passage

45 of time and with repeated remembering (Lifanov et al., 2021). Repeated retrieval of naturalistic
46 events over the course of a week has been found to stabilize each person’s recall (for example,
47 which details they include) without causing different people’s recalls to converge (Melega et al.,
48 2025). Taken together, these observations suggest that each time we remember an experience, the
49 substance of the underlying memory we are retrieving may differ from what we retrieved before.

50 How might we compare two recountings of the same experience when the experience and
51 both recountings take the form of unconstrained natural language? Recent work has begun to use
52 language models to quantify the content of what individuals say when they recount what they
53 remember (Fenerci et al., 2025; Fitzpatrick et al., 2026; Heusser et al., 2021; Manning et al., 2022).
54 The approach developed in our prior work embeds the description of an experience, and each
55 individual’s recounting of it, into a common high-dimensional space whose coordinates reflect
56 the concepts under consideration, and through which the experience and recountings each unfold
57 over time as a geometric trajectory (Fitzpatrick et al., 2026; Heusser et al., 2021; Vera et al., 2025).
58 We then segment the experience and each recounting into events and identify which event from
59 the experience each recall event describes, yielding sequences of events that may be compared
60 directly across people and across delays (Baldassano et al., 2017; Chen et al., 2017; Heusser et al.,
61 2021).

62 Here we apply our approach to data from an experiment in which participants viewed an
63 episode of a television show, recounted its events immediately afterward, and then recounted
64 them again following a week-long delay. Because we collected two recountings of the same
65 experience from each participant, we are able to ask what participants’ delayed recountings most
66 closely resemble. If retrieval recovered the original experience in an uncorrupted form, then both
67 recountings should be about equally similar to the experience itself. We find instead that the
68 structure of a participant’s delayed recounting is more similar to that of their own immediate
69 recounting than to that of the original episode. It is also more similar to their own immediate
70 recounting than to other participants’ immediate recountings, to the degree that an individual’s
71 immediate recall can be identified from their delayed recall on the basis of this structure alone.
72 What governs retrieval after a delay therefore appears to be a person-specific distillation, rather

73 than the episode as it was first experienced.

74 Quantifying the semantic geometry of individuals' recalls at different points in time offers the
75 opportunity to characterize the transformations undergone by our memories beyond an initial
76 recounting. For example, if individuals' recountings tend to retain their idiosyncratic structures
77 after a delay, how *do* the structures of our memories tend to change with time? There is a particular
78 qualitative change our memories undergo as we naturally forget aspects of experience, implied by
79 common descriptors such as "fading", "blurring", becoming "fuzzy", or "decaying" (Thorndike,
80 1913). But the nature of this change can be difficult to characterize more concretely. Here we show
81 that this transformation is well described as geometrically simplifying the path through semantic
82 space an individual follows as they recount an experience, in that applying existing algorithms
83 for removing detail from a geometric curve to participants' immediate recalls identifies the details
84 they are likely to forget following the delay.

85 Results

86 *[OUTLINE — placeholder structure, not final prose. Bracketed italic text marks content still to*
87 *be written. No statistic below has been computed or verified; every "report [STAT]" is a hole to be*
88 *filled from the cited notebook's output.]*

89 *[FIGURE BUDGET: 8 main-text figures maximum. The plan below uses 6, leaving 2 in re-*
90 *serve (one likely wanted for the RDP analysis if it needs a second panel set, one for contin-*
91 *gency). Figure 1 (**paradigm.pdf**) is the only finished main-text figure; the other PDFs already in*
92 ***paper/figures/source/** are analysis outputs that may feed panels rather than figures in their*
93 *own right. Nothing else is made yet.]*

94 From episode and recalls to comparable event sequences

95 *[Walk the reader through the pipeline, briefly and conceptually (details live in the Methods), up to the point*
96 *where we have three things in hand: (a) the sequence of events comprising the episode, (b) each participant's*
97 *sequences of immediate and delayed recall events, and (c) the mapping between each recall event and the*

98 episode event it describes. This is the setup that everything below depends on — the point to land is that the
99 episode and both recalls now live in a common space and are segmented into comparable units.]

100 [Figure 1 — **paradigm**. `paradigm.pdf`, already in `paper/figures/source/`. Session-1 row: in-
101 structions, episode (24 min), immediate recall (10–60 min), prediction (5 min); one-week gap; session-2 row:
102 instructions, delayed recall (10–60 min), session-2 episode (20/22 min), recall of that episode (10–60 min).
103 Planned revision: reduce the opacity of the blocks for the phases we don’t analyze — the session-1 prediction
104 block, and the session-2 episode-viewing and subsequent-recall blocks — so the reader’s eye lands on the
105 immediate/delayed recall pair that the paper is actually about.]

106 [Figure 2 — **methods overview**. To be made. Annotations and recall text → trajectories → event-
107 segmentation procedure → event sequences. Loosely modeled on Fig. 2 of Heusser et al. (2021), extended
108 with the (new) event-segmentation step. Layout still open. Candidate source material already generated:
109 `episode-corrmat.pdf` and `k-optimization-episode.pdf` (from `event-segmentation.ipynb`), and
110 the per-recall similarity heatmaps and match matrices `atlep1-and delayed-recall-similarity-heatmaps.pdf`
111 `/ -matchmats.pdf`.]

112 **Participants recall the episode’s events in a similar order before and after the** 113 **delay**

114 [Motivation: we wanted to know whether participants remembered the episode similarly before and after the
115 delay. The simplest way to ask this is whether they recalled the same events in the same order.]

116 [Result: recall-order similarity. Computed in `self-similarity-analyses.ipynb`, section “Is de-
117 layed recall order more similar to own immediate recall order than to others’ immediate recall orders?”. The
118 measure compares pairwise precedence relations over the episode events shared between two recalls, then
119 percentile-ranks each participant’s delayed-vs-own-immediate similarity against their delayed-vs-others’-
120 immediate similarities, tested against chance (0.5) with a Wilcoxon signed-rank test. Report [STAT].]

121 [Figure 3 — **recall-order similarity**. To be made; multi-panel.]

122 [Panel A: illustration of the method. For one example participant, show the matrix of pairwise recall-
123 order-precedence values over pairs of episode events — probably just the lower triangle — for their delayed

124 recall beside their own immediate recall. Grey out (or reduce the opacity of) cells for event pairs not recalled
125 in both sessions, since the comparison is restricted to pairs present in both. Then show the same thing again
126 for that participant's delayed recall against a different participant's immediate recall, which is the contrast
127 the analysis actually tests. This maps directly onto `pairwise_precedence()` and the `shared_events`
128 argument to `order_similarity()` in the notebook, so the panel and the code should be checked against
129 each other.]

130 [Panels B–D: de-slopified versions of panels B–D of the reference mock-up. Include the analogues of that
131 mock-up's B and D so that the same panel types can reappear in the recall-structure figure (Figure 5), letting
132 the reader compare order and structure on identical terms — this parallelism is the reason for the pairing
133 and should be preserved in the final layout.]

134 [Note on ordering: this is the least novel analysis of the three and the one least dependent on our
135 framework — order could be measured without any embedding model. It goes first precisely because it
136 establishes the basic effect on the simplest possible terms, setting up the contrast with what follows.]

137 **The semantic structure of participants' recalls is preserved across the delay**

138 [Motivation: order is only the coarsest description of a recall. Because our approach embeds recalls in a
139 continuous semantic space, we can ask a stronger question — whether the structure of how a participant
140 moved through the episode's content is preserved, not merely the sequence in which they hit the events.]

141 [Figure 4 (optional) — 2D trajectories. Decision deferred, and it depends on how this section's
142 narrative settles. Argument for including it: it introduces the idea of geometric comparison and establishes
143 the 2D paths before they get used as examples in Figure 5, so the reader isn't meeting the representation
144 and the result at the same time. Argument against: it spends one of 8 figure slots on setup. Middle
145 option — a compact version showing only the episode path plus immediate and delayed paths for a handful
146 of participants, with the full version relegated to the supplement. Source: `2d-trajectories.ipynb`
147 (`atlep1-paths2d.pdf` and `delayed-paths2d.pdf`, `imm-del-2d.pdf`); note those `savefig` calls are
148 currently commented out in the notebook.]

149 [Result: trajectory/event-structure similarity. Computed in `self-similarity-analyses.ipynb`,
150 sections "Is delayed recall more similar to own immediate recall than to episode?" and "Is delayed recall

151 *more similar to own immediate recall than to others' immediate recalls?''*. Both use DTW warp cost (average
152 *cosine distance between matched events*) over mean-centered event sequences; the second percentile-ranks
153 *self against others and tests against chance. Report [STAT] for each.*]

154 *[Framing: this result appears strong enough to present as a "fingerprint" — i.e., that one can identify*
155 *which immediate recall belongs to a given participant from the structure of their delayed recall alone. Frame*
156 *it that way in both the text and the figure, but confirm the effect size supports the claim before committing*
157 *to that language.]*

158 **[Figure 5 — recall-structure similarity.** *To be made; the largest figure in the paper, covering both*
159 *analysis (A) "delayed is more similar to immediate than to the episode" and analysis (B) "delayed is more*
160 *similar to own immediate than to others' immediates". Layout still open.]*

161 *[Example trajectory panels (3, from the current draft): the left panel is common to both analyses, so*
162 *it should sit centrally in the final arrangement; the right panel is the contrast for analysis (A) and the*
163 *middle panel is the contrast for analysis (B). Planned refinements: thicker lines; faint connectors between*
164 *DTW-matched events within each panel (already present in the draft); possibly PCHIP splines rather than*
165 *straight segments between points, to make the paths read as curves. Under consideration — a fourth panel*
166 *showing pairwise mean point-by-point distances between trajectories, which would require upsampling each*
167 *trajectory to a common number of points and comparing them in order (or substituting DTW cost, if that*
168 *turns out to read better).]*

169 *[Analysis (A) panel: per-participant connected dots showing each participant's delayed-to-immediate*
170 *versus delayed-to-episode alignment cost. Probably without the KDEs from the draft version.]*

171 *[Analysis (B) panels (3): de-slopified versions of the current draft. Two of them are the direct analogues*
172 *of Figure 3's panels B and D and show stronger effects — keeping the panel types visually identical across*
173 *the two figures is what makes "structure beats order" legible without the reader doing arithmetic. The third*
174 *gets at how person-specific the similarity is, and is the panel that carries the "fingerprint" claim.]*

175 *[Candidate supporting material: `atlep1-recall-corrstats.pdf` and `delayed-recall-corrstats.pdf`*
176 *(from `event-segmentation.ipynb`).]*

Delayed recalls are simplified versions of participants' immediate recalls

[Motivation: having established that a participant's delayed recall structure resembles their own immediate recall structure, the natural next question is how it changes over the delay.]

[Result: trajectory simplification. Computed in `rdp-analyses.ipynb`. Each participant's immediate recall trajectory is simplified with the Ramer-Douglas-Peucker algorithm over a range of tolerances (ϵ), and the ϵ minimizing DTW warp cost to their delayed recall trajectory is selected. Sub-results to report: (i) the relationship between how many events a participant dropped across the delay and how many RDP removes at the best-fitting ϵ (Kendall's τ); (ii) the same against ϵ itself; (iii) whether RDP predicts which events were forgotten, benchmarked against a hypergeometric null and compared to predictions derived from other participants' trajectories. Report [STAT] for each.]

[Figure 6 — RDP simplification. To be made, and how best to illustrate the simplification is genuinely unresolved — flagged as an open design question rather than a layout still to be drawn. The regression plots for sub-results (i) and (ii) exist in `rdp-analyses.ipynb` but have no `savefig` calls, so nothing is exported to `paper/figures/source` yet. One thing worth trying: since Figure 5 will already have taught the reader to read 2D trajectories, showing one participant's immediate path beside its RDP-simplified version and their actual delayed path would make “the delayed recall looks like a simplified version of the immediate one” visible directly, with the quantitative panels as support. If the analysis needs a second panel set, one of the 2 reserve figure slots can absorb it.]

[@paxton: does the simplification story want a fourth subsection on what the dropped events have in common (e.g. are they lower-salience, more peripheral to the narrative), or is that a Discussion point?]

Discussion

We asked participants to watch an episode of a television show, to recount what they remembered immediately afterwards, and to recount the episode again following a week-long delay. Embedding the episode's annotations and each recounting into a common semantic space allowed us to represent all three as trajectories, to segment each trajectory into events, and to compare them on common terms. Participants recalled the episode's events in a similar order before and after the

203 delay. The semantic structure of a participant's delayed recall was more similar to the structure
204 of their own immediate recall than to that of the episode, and it was more similar to their own
205 immediate recall than to other participants' immediate recalls. Delayed recalls were also well
206 described as simplified versions of immediate recalls, in that removing detail from a participant's
207 immediate recall trajectory brought it closer to the delayed recall trajectory that followed it.

208 Taken together, these results suggest a way of thinking about what recounting an experience
209 does to it. If experiences and memories may be described as trajectories through a semantic
210 space, then a recounting is a traversal of one of those trajectories (Heusser et al., 2021). Semantic
211 space differs from physical space in that nothing prevents a narrator from jumping between distant
212 locations, and yet participants' recountings turn out to be strongly autocorrelated in practice, which
213 is the regularity that our event-segmentation approach relies on. Because movement through
214 the space is relatively continuous, describing how one arrives at a later event from an earlier
215 one implies that the intervening region was traversed in some sense, whether it was articulated
216 explicitly or left for the listener to fill in. Some events therefore behave like control points, in that
217 removing one substantively changes the shape of the path. Others behave like waypoints, which
218 fall close to the line between their neighbors and may be removed with little consequence, since
219 travelling between the surrounding control points already implies passing near them. Trajectory
220 simplification provides a way of telling the two apart, and this distinction appears to hold at both
221 of the transitions we can examine. The events that simplification removes from the episode's own
222 trajectory are the events that the fewest participants recounted immediately afterwards (**[VERIFY**
223 **THIS]**), and the events that simplification removes from a participant's immediate recall trajectory
224 are the ones that participant tended to drop across the delay.

225 That not all events are equally memorable is well established. Gist and central details, for
226 example, survive delays that perceptual and peripheral details do not (Gilboa and Moscovitch,
227 2021; Heusser et al., 2021; Sekeres et al., 2016, 2018), and autobiographical narratives shed some
228 classes of content faster than others (Diamond and Levine, 2020; Su et al., 2026). Memories also
229 become more semantic and less perceptually detailed with the passage of time and with repeated
230 remembering (Lifanov et al., 2021), and events with stronger connections to the events around them

231 are better remembered than more isolated ones (Antony et al., 2024; Lee and Chen, 2022). What our
232 results add is a change in where that asymmetry comes from. Centrality is ordinarily established
233 from the outside, by asking raters which details matter to the plot, or by sorting content into
234 categories fixed in advance. Here it follows from the geometry of an individual's own trajectory,
235 and it is computable from that trajectory alone. An event is central because of where it sits relative
236 to the rest of the path, and not because a rater judged it to be important. This turns the observation
237 that some events are more memorable than others into a candidate explanation of why.

238 Why should waypoints be the events that go? One possibility is that they are redundant,
239 in the sense that a listener, or the rememberer's own future self, can recover them from the
240 events on either side. Another is that they are simply weaker in themselves, for example less
241 distinctive, less emotionally charged, or less causally connected to the rest of the narrative (Antony
242 et al., 2024). These accounts make similar predictions about which events will be forgotten, but
243 they differ in what those predictions should depend on. If events are dropped because they are
244 intrinsically weak, then a simplification derived from any participant's trajectory should predict
245 forgetting about as well as one derived from the participant's own, since weak events would be
246 weak for everyone. If they are dropped because they are recoverable from their neighbors, then
247 a participant's own trajectory should do better, since control-point status is defined relative to the
248 rest of that person's path. Our results favor the latter account. Read this way, dropping a waypoint
249 is an economy rather than a loss, and it is worth noting that events which go unreported are
250 not always unavailable, since a shift in perspective at retrieval can recover details that an earlier
251 recounting left out (Anderson and Pichert, 1978; Oedekoven et al., 2017).

252 The same argument implies that what counts as a control point should be idiosyncratic. An
253 event's status depends on the path it sits within, and different people who watched the same
254 episode construct different paths through it, so the events whose removal would deform one
255 person's trajectory need not be the events that matter for another's. This is what we take the
256 specificity of the delayed recalls to mean. That a delayed recall can be matched to the immediate
257 recall which preceded it, on the basis of structure alone, is a fingerprint in the sense that the term is
258 used for functional connectivity and for other individual signatures measured during naturalistic

259 experience (Anderson et al., 2020; Finn et al., 2020, 2015). People differ, for example, in where they
260 place event boundaries in a continuous experience (Sava-Segal et al., 2023), in how much of their
261 recall structure they share with others (Chen et al., 2017; Nguyen et al., 2019), and in the character
262 of their autobiographical remembering more generally (Palombo et al., 2018). Repeated retrieval
263 of naturalistic events has been found to stabilize each person's recall without causing different
264 people's recalls to converge (Melega et al., 2025), which is what one would expect if each person is
265 stabilizing their own set of control points.

266 Everything we measured came from a verbal recounting, and so a reasonable objection is that
267 control points and waypoints are properties of storytelling rather than of memory. We think that
268 dichotomy is less clean than it first appears. Recounting is the only observable available in a study
269 of this kind, so the two are confounded by construction, and no analysis of transcripts alone can
270 fully separate them. More to the point, the way a person tells a story is not merely a window onto
271 what they remember, but also an input to what they will remember next. Retelling an event for a
272 particular audience or purpose, for example, biases how it is recalled later (Marsh, 2007; Tversky
273 and Marsh, 2000), and reproducing a narrative repeatedly reshapes it in ways that persist (Bergman
274 and Roediger, 1999; St Jacques and Schacter, 2013). If the function of recounting is to let a listener,
275 or one's own future self, re-traverse the trajectory of the original experience, then communicative
276 efficiency and mnemonic efficiency are not competing explanations so much as the same pressure
277 seen from two sides.

278 We acknowledge several limitations in the current study. First, every participant recalled the
279 episode immediately, and so we cannot separate two accounts of what we observed. A delayed
280 recall may resemble the immediate recall because the immediate recall was itself re-encoded and
281 then retrieved a week later, or because a single stable memory was sampled twice and each
282 sampling was shaped by the same person-specific distillation. Our measures are symmetric and
283 cannot establish direction, and the design lacks an arm in which the immediate recall is withheld.
284 Designs that do include such a control, in which retrieved information is later found nearer
285 to what was retrieved than to what was originally presented, offer the strongest evidence that
286 retrieval writes its own output back into the trace (Bridge and Paller, 2012; Nader and Hardt, 2009;

287 Nader et al., 2000). Second, we used one episode, one delay, and one intervening recounting, and
288 it remains to be seen how the picture changes across stimuli, across longer intervals, or across
289 many successive retellings (Chan et al., 2009; Lifanov et al., 2021). Third, control-point status is
290 defined with respect to one particular simplification algorithm, and a different algorithm might
291 well partition the same events differently. Fourth, our procedure matches every recall event to
292 an event in the episode, so content that a participant introduced but that does not appear in the
293 episode falls outside what we are able to measure.

294 Speculatively, this account suggests what might become of a memory that is recounted many
295 times. If each retelling preserves control points and sheds waypoints, and if each retelling then
296 becomes the trajectory that the next one simplifies, a memory might converge over many retellings
297 toward a skeleton of control points, with the intervening detail increasingly implied rather than
298 stated. Whether that convergence has a floor, and whether the surviving skeleton continues to
299 resemble the original experience or drifts steadily away from it, are questions our data cannot
300 answer. The possibility seems worth taking seriously outside the laboratory, since the settings
301 in which people are asked to recount an experience repeatedly are often the settings in which
302 accuracy matters most. An eyewitness who has already described what they saw, for example, is
303 describing a description the next time they are asked (Chan et al., 2009). A patient recounting a
304 formative episode across successive sessions may be reshaping it in the process, and a teacher who
305 explains an idea the same way each term may be simplifying it past the point where it still carries
306 what made it worth explaining.

307 Ultimately, our work points toward a set of questions about how the geometry of a recounting
308 relates to the experience that it describes. Do the events that behave like control points in a
309 recounting correspond to the boundaries at which neural activity shifts during the experience
310 itself (Baldassano et al., 2017; Chen et al., 2017)? Can they be predicted in advance from the
311 stimulus, or are they recoverable only after the fact, from a particular person's path? Does a listener
312 who hears a recounting actually re-traverse the speaker's trajectory, filling in the waypoints that
313 the speaker left out? And because our measure of what connects one event to another is built from
314 semantic similarity, it cannot on its own distinguish events that are close in meaning from events

315 that are causally linked (Antony et al., 2024; Lee and Chen, 2022; Xu et al., 2024). Collecting causal
316 and importance judgments for this episode would separate geometric control-point status from
317 causal centrality, and would tell us whether these two notions of what makes an event matter are
318 one notion arrived at by different routes.

319 **Methods**

320 Our experimental protocol was approved by the Committee for the Protection of Human Subjects
321 at Dartmouth College. We obtained written informed consent from all participants prior to the
322 start of both experiment sessions.

323 **Participants**

324 We enrolled a total of 70 Dartmouth undergraduate students in our study. Participants received
325 optional course credit for enrolling and were pre-screened to ensure they had not previously seen
326 any episode of either TV show viewed during the experiment. No statistical method was used
327 to predetermine sample size. Prior to data processing and analysis, we excluded a total of 17
328 participants from our dataset: 4 encountered technical issues with the experiment, 2 self-reported
329 difficulty with the task due to illness, 4 spoke for less than 10 minutes during the immediate recall
330 phase, and 7 did not return for the second experiment session. The data we analyzed comprised
331 the remaining 53 participants.

332 At the start of the experiment, we asked each participant to complete a demographic survey
333 that included questions about their age, sex, race, and ethnicity. Participants' ages ranged from 18
334 to 22 years (mean: 18.91 years; standard deviation: 1.01 years). A total of 41 participants reported
335 their sex as female and 12 participants reported their sex as male. Participants reported their races
336 as White (34 participants), Asian (16 participants), Black or African American (3 participants),
337 American Indian or Alaska Native (1 participant), and other (1 participant). (Note that some
338 participants selected multiple racial categories.) A total of 52 participants reported their ethnicity
339 as "Not Hispanic or Latino" and 1 participant reported their ethnicity as "Hispanic or Latino."

340 We also asked participants their native spoken language; whether they had any hearing, speech,
341 or color vision impairments; and about any current medications or recent injuries that could impact
342 attention or memory. All participants reported that their native spoken language was English, that
343 they had no hearing or speech impairments, and that they had normal color vision. A total of 52
344 participants reported no attention- or memory-relevant medications or injuries. One participant
345 reported sustaining a concussion more than 3 years prior to the experiment and was not excluded
346 from our dataset.

347 Additionally, before each of the two experiment sessions, we asked participants about their
348 sleep, coffee consumption, and level of alertness. Participants reported having slept between 5
349 and 11 hours the night prior to session 1 (mean: 7.37 hours; standard deviation: 1.45 hours) and
350 between 5 and 11 hours the night prior to session 2 (mean: 7.39 hours; standard deviation: 1.23
351 hours). They reported having consumed between 0 and 3 cups of coffee on the day of session
352 1 (mean: 0.43 cups; standard deviation: 0.72 cups) and between 0 and 3 cups of coffee on the
353 day of session 2 (mean: 0.48 cups; standard deviation: 0.68 cups). Participants reported their
354 current levels of alertness on a 5-point Likert scale. At the start of session 1, 1 participant reported
355 feeling "Very sluggish," 14 reported feeling "A little sluggish," 18 reported feeling "Neutral," 15
356 reported feeling "A little alert," and 5 reported feeling "Very alert." At the start of session 2, 3
357 participants reported feeling "Very sluggish," 9 reported feeling "A little sluggish," 23 reported
358 feeling "Neutral," 17 reported feeling "A little alert," and 3 reported feeling "Very alert."

359 *@jeremy: do we want to report class years and majors? And/or do we want to report post-experiment*
360 *survey responses (how engaging did you find the episode, how easy/difficult was it to follow, how well do you*
361 *think you recalled it, how well do you think you learned characters' names)?*

362 Experiment

363 The experiment took place over two sessions lasting approximately one hour each, separated by
364 6–8 days (mean: 6.76 days; standard deviation: 0.73 days). Prior to the start of the experiment,
365 participants were assigned to one of two groups (A or B), alternating based on enrollment order.
366 During session 1, all participants first viewed the pilot episode of the FX series *Atlanta* ("The Big

367 Bang”; duration: 24 min 15 s). Immediately after viewing, participants were given up to one hour
368 to recount what happened in the episode to the best of their abilities. Following Chen et al. (2017),
369 participants were instructed to describe the episode in as much detail as possible and to try to
370 speak for a minimum of 10 minutes. They were told to try to recount the episode’s events in their
371 original order, but that they were free to go back and describe earlier events should they realize
372 they had missed something. After the immediate recall phase, participants were then given up to
373 5 minutes to predict what they thought would happen in the show’s next episode.

374 Upon returning for session 2, all participants were again given up to one hour to recount the
375 *Atlanta* episode they had viewed the week prior. Following the delayed recall phase, participants
376 in Group A then viewed the second episode of *Atlanta* (“Streets on Lock”; duration: 21 min
377 57 s) while participants in Group B instead viewed the pilot episode of the FOX series *Arrested*
378 *Development* (“Pilot”; duration: 20 min 37 s). Both groups were then given up to one hour to
379 recount the just-watched episode in the same manner and given the same instructions as in session
380 1. We implemented the experiment using the psiTurk platform (Gureckis et al., 2016) and jsPsych
381 library (de Leeuw, 2015), with a custom jsPsych plugin we developed in prior work for recording
382 verbal recall audio (Ziman et al., 2018). Here we analyze participants’ immediate and delayed
383 recalls of the *Atlanta* episode viewed in session 1.

384 **Stimulus annotation and recall transcription**

385 To capture the evolving semantic content of the *Atlanta* episode, we created detailed text annota-
386 tions of the on-screen content during each individual camera shot throughout the episode (mean
387 annotation duration: 3.05 s; standard deviation: 1.80 s). Specifically, we used ANVIL (Kipp, 2001)
388 to record, for each camera shot, a set of semantically relevant features. These included a brief de-
389 scription of physical occurrences and characters’ explicit actions, a brief description of characters’
390 internal states (e.g., emotions, motivations) and other implicit content, the names of all characters
391 appearing on screen, a transcription of any dialogue, the name(s) of the character(s) speaking, the
392 shot’s setting, whether the shot took place indoors or outdoors, and whether or not there was
393 music playing during the shot. This feature set was adapted from a similar annotation corpus

394 created by Chen et al. (2017), which we used in developing our trajectory-analysis framework in
395 prior work (Heusser et al., 2021).

396 We then transcribed each participant’s immediate and delayed verbal recall of the *Atlanta*
397 episode. Our transcription process consisted of three steps: an initial manual transcription, a
398 two-pass prose-formatting conversion with a large language model, and a final manual cleanup
399 and correction step. First, we manually transcribed each recall audio file collected during the
400 experiment. These initial transcripts reflected participants’ spoken recalls verbatim, without any
401 punctuation or additional formatting, and including disfluencies, stuttered repetitions, and other
402 artifacts common to unrehearsed speech. However, since transformer models (such as we used
403 to obtain text embeddings of participants’ recalls) are trained almost exclusively on written text,
404 structural conventions like sentence breaks, punctuation, and capitalization provide an important
405 signal these models leverage to contextual other tokens (e.g., words) and construct a representation
406 of the text’s overall meaning. Meanwhile, speech-native phenomena such as stuttered repetitions
407 and filler words (“like”, “um”, etc.) that are comparatively rare in written text contribute out-of-
408 distribution noise that can be expected to degrade embedding fidelity, and prior work has shown
409 that automated removal of these phenomena improves performance on downstream tasks that
410 assume clean, written-style input (Chen et al., 2022; Wang et al., 2010). We therefore used a large
411 language model (Qwen3.5 122b; Qwen Team, 2026) to convert participants’ recall transcripts to
412 a structure and style more typical of written text. We used the `langchain_dartmouth` Python
413 package (Stone et al., 2025) to implement this conversion as two sequential passes over each
414 transcript. In the first pass, we prompted the model to split the transcript into sentences at
415 natural breaking points, add punctuation including quotes around reported speech, remove clear
416 non-word disfluencies (“um,” “uh,” “hmm,” etc.), collapse stuttered verbatim repetitions, and
417 mark self-corrections and restarts with em-dashes. In the second pass, we prompted the model
418 to remove all instances of “like” used as a filler word. We removed filler “like” instances in a
419 separate pass because of their high frequency of occurrence in the transcripts and the additional
420 effort to required to distinguish them from meaningful uses of “like.” The full texts of our prompts
421 for both passes can be found in *Supplementary methods*. Finally, we manually reviewed all LLM-

422 formatted transcripts to identify and correct any errors, ensure sentence breaks were appropriate
423 and consistent across participants, confirm that no content was improperly added to or removed
424 from the transcripts, and resolve any ambiguities based on the prosody of participants’ original
425 recall audio.

426 **Analysis**

427 **Statistics**

428 Wilcoxon tests were one-sided because our alternative hypothesis was directional (delayed matches
429 own immediate *better than* the episode, others’ immediates, etc.).

430 **Constructing episode and recall trajectories**

431 We adapted a method we developed in prior work (Heusser et al., 2021) to transform the *Atlanta*
432 episode and participant’s recalls of it into trajectories through a common representation space
433 using a text embedding model. Briefly, text embedding models take in natural language text
434 of an arbitrary length and output a fixed-length vector representation that captures the text’s
435 semantic meaning. Intuitively, an output vector of length n can be viewed as a coordinate in an
436 n -dimensional text embedding space. While specific model architectures and training paradigms
437 vary, modern text embedding models are broadly trained to output similar coordinate vectors (in
438 terms of cosine similarity, dot product, inverse Euclidean distance, or other geometric measures) for
439 semantically similar input texts, even when those texts share few or no specific words. We obtained
440 text embeddings for the episode and recalls using EmbeddingGemma (Vera et al., 2025), a pre-
441 trained 308-million-parameter encoder-only transformer model that outputs coordinate vectors in
442 a 768-dimensional embedding space.

443 To construct the trajectory for the *Atlanta* episode, we first converted our tabular annotations
444 of the content present during each camera shot to a plain-text format. Separately for each shot,
445 we concatenated the annotation text describing explicit on-screen occurrences and characters’
446 implicit internal states. We then appended “Speech: ” followed by the shot’s dialogue, “Character

447 speaking: " followed by the name of the speaker, and "Setting: " followed by the shot's setting. If
448 any annotated fields were empty for a given shot (e.g., if no character was speaking at that time), we
449 omitted them and their prefixes from the concatenated text. Next, we used EmbeddingGemma to
450 transform the text snippet for each camera shot into a text embedding vector, resulting in a number-
451 of-shots (478) by number-of-features (768) matrix. Since the camera shots varied in duration, we
452 then resampled this series of embedding vectors to a consistent temporal resolution. We assigned
453 each vector to the timepoint midway between the start and end of its corresponding camera shot
454 (in seconds), and used linear interpolation to estimate a vector for each second of the episode. This
455 yielded a number-of-seconds (1,455) by number-of-features (768) matrix—i.e., a 768-dimensional
456 *trajectory*.

457 We then created similar trajectories for each participant's immediate and delayed recall of the
458 episode. We began by splitting each recall transcript into a list of sentences, with the constraint
459 that splits could not occur within reported speech (i.e., if a participant recounted a multi-sentence
460 quote by a character in the episode, we treated the full quote as a single sentence-unit). We
461 then parsed these sentences into overlapping sliding windows that spanned up to 5 consecutive
462 sentences each. These windows grew to and shrank from their full size at the beginning and end of
463 a participant's recall sequence—in other words, the first window contained only the first sentence
464 of the participant's recall, the second contained the first two sentences, and so on. This ensured
465 that all sentences from a participant's recall transcript appeared an equal number of times (5) in
466 their sequence of sliding windows. Finally, we used EmbeddingGemma to transform each sliding
467 window into a text embedding vector. Repeated for all recall transcripts, this yielded a separate
468 trajectory characterizing each participant's recall from each of the two experiment sessions. The
469 number of samples in participants' recall trajectories (i.e., the number of sliding windows parsed
470 from their recall transcripts) ranged from 61 to 308 (mean: 136.68; standard deviation: 49.70).

471 **Relating episode and recall embedding vectors**

472 In prior work using our trajectory-analysis framework (e.g., Fitzpatrick et al., 2026; Heusser et al.,
473 2021; Manning, 2021; Manning et al., 2022; Raccach et al., 2024), embeddings of the stimulus and

participants’ responses were constructed using a topic model (Latent Dirichlet Allocation; Blei et al., 2003) trained solely on the stimulus content. Here we elected to instead use a pre-trained transformer-based text embedding model due to the richer and more nuanced textual information these models are able to capture. (For example, these context-sensitive models can disambiguate the texts “Earn looked at Alfred” and “Alfred looked at Earn” in their representations, whereas bag-of-words models like LDA would represent the two identically.) However, these richer representations also introduce certain geometric nuisances we needed to account for in order to effectively relate the contents of participants’ recalls to that of the episode they viewed. First, because modern transformer models are pre-trained on large and diverse text corpora, the representational capacities of their embedding spaces span an enormous breadth of semantic content. Consequently, in preliminary analyses we found that the embeddings of our annotations and participants’ recalls (which were all “about” the events, themes, and characters of a single *Atlanta* episode) were strongly anisotropic—i.e., they all occupied the same relatively narrow angular region of the model’s overall embedding space. Additionally, because transformer models are sensitive to the syntactic structure of text, their embedding representations carry latent information about the text’s linguistic register in addition to its semantic content. Accordingly, we found there was a consistent angular offset between embeddings of the episode annotations and those of participants’ recalls, reflecting the fact that the annotations comprise grammatically well-structured written text while the recalls comprise transcripts of casual unrehearsed speech.

We corrected for these representational artifacts by mean-centering the episode and recall trajectories before performing analyses that involved measuring similarities between embedding vectors within or across trajectories. Conceptually, mean-centering a given set of embedding vectors (i.e., subtracting the across-vector average from each individual vector) serves to remove representations that are common across all vectors under consideration, and instead highlight representations that *vary* between them. For example, consider the sequence of embedding vectors comprising the trajectory of the *Atlanta* episode. Common across vectors within this sequence will be latent signals driven by how we chose to format the annotation text passed to the embedding model, the annotator’s particular idiolect (i.e., idiosyncratic vocabulary and writing style), and any high-level

502 overarching themes of the episode. *Not* common across these vectors are the finer-grained semantic
503 representations that distinguish different events, scenes, and actions throughout the episode from
504 each other. We found that removing these non-discriminative signals (to the extent they behave as
505 additive offsets) via mean-centering helped reduce the embeddings’ anisotropy (i.e., increased the
506 range of measured between-vector similarities) and improved our ability to segment the episode
507 trajectory into meaningful events. Similarly, mean-centering both the episode and recalls before
508 relating their embedding representations helped mitigate broad-scale representational differences
509 driven by their diverging registers, and enabled us to better relate them based on event-specific,
510 content-driven patterns.

511 Our approach to mean-centering the recalls was guided by the particular latent representation
512 we wanted to remove versus retain. In segmenting each participant’s recall trajectory from each
513 session into events (see *Event-segmentation*), our goal was to discriminate between embedding
514 vectors strictly within a single recall sequence in order to identify temporally transient content-
515 related patterns. We therefore centered each recall trajectory (in addition to the episode) on its own
516 centroid to remove non-discriminative representations (e.g., register, broad-scale thematic content)
517 and maximize the contrast between vectors within that single sequence. However, the analyses
518 shown in Figures **REF TO DTW FIG** and **REF TO RDP FIG** entailed comparing sequences of
519 recall events not only to the episode, but also to each other, both within- and across-participants, to
520 examine whether within-participant similarities tended to exceed across-participant similarities.
521 Here our goal was to remove representational artifacts that could work to bias our results towards
522 significance. Specifically, participants’ idiolects (i.e., particular speaking styles and sets of words
523 with which they describe the episode) will influence the embedding representations of their recalls
524 in ways that tend to be similar between their immediate and delayed recalls, but different across
525 participants. To correct for this confound, we jointly centered each participant’s set of immediate
526 and delayed recall events on the two sequences’ shared centroid (i.e., the average vector across
527 all events from both sessions). This served to reduce the recall embeddings’ anisotropy (by
528 removing signals driven by high-level themes of the episode they recalled) and mitigate the effects
529 of the spoken-recall register and participant-specific idiolect (both of which are shared between a

530 participant’s recalls from the two sessions) while preserving any genuine drift in the high-level
531 content of each participant’s recall over the delay (which per-session mean-centering would instead
532 discard).

533 **Event-segmentation**

534 After constructing trajectories for the episode and each participant’s immediate and delayed recall,
535 we used hidden Markov models (HMMs; Baum and Petrie, 1966; Rabiner, 1989) to segment each
536 trajectory into a sequence of discrete events. Given a number of states K , an HMM models an
537 observed time series as a progression through K hidden states, each of which has a characteristic
538 signature in the data (i.e., a pattern that timepoints belonging to that state tend to exhibit) and
539 tends to persist across timepoints with occasional transitions rather than fluctuating freely. Because
540 states persist, inferring the most probable state at each timepoint segments the observed time series
541 into a sequence of events—i.e., spans of contiguous timepoints with the same most-probable state.

542 We segmented the episode trajectory into events using the HMM implemented in the BrainIAK
543 toolbox (Kumar et al., 2021), which jointly learns a mean pattern for each of K states and the set
544 of transitions between them via the Baum-Welch algorithm (Baum et al., 1970), with the added
545 constraint that each state is visited exactly once in a strict left-to-right order (Baldassano et al., 2017).
546 To select an appropriate number of states (i.e., events) for the episode, we used an optimization
547 procedure we developed in prior work (Heusser et al., 2021) that aims to identify the single
548 timescale of event representations emphasized most heavily in the episode’s narrative structure.
549 We began by segmenting the (mean-centered) episode trajectory into K events for each K between
550 2 and 50, inclusive. For each candidate segmentation, we then computed the cosine similarity
551 between each pair of timepoints assigned to the same event, and between each pair of timepoints
552 assigned to different events. When constructing these distributions of within- and across-event
553 similarities, we restricted comparisons to pairs of timepoints separated by no more than the
554 duration of the longest event in the candidate segmentation, so that the within- and across-
555 event distributions were computed over comparable temporal distances and the across-event
556 distribution was not dominated by distant timepoint pairs that we would expect to be dissimilar

557 regardless of event structure. We then selected the candidate segmentation that maximized the
558 Wasserstein distance between these within- and across-event similarity distributions. We found
559 that the maximally distinctive and internally self-similar segmentation was yielded by $K = 20$
560 events. **(TODO: ADD SUPP FIG SHOWING WASSERSTEIN DIST AS FUNCTION OF K)**
561 Finally, we averaged the embedding vectors over timepoints between each pair of adjacent event
562 boundaries to obtain a sequence of stable event-average representations for the episode.

563 To segment participants’ recall trajectories into sequences of events, we developed a novel
564 HMM-driven procedure that grounded the recalls directly against the content of the episode
565 participants viewed and recounted. Essentially, rather than re-learning separate sets of state
566 representations from each participant’s recalls (which tend to be shorter, noisier approximations
567 of the episode content), we leveraged the state representations already learned from the episode to
568 identify transitions between the episode events a given participant was (most probably) describing
569 at a given point in time. First, we mean-centered the sequence of event-average embedding
570 vectors for the episode, and each participant’s sequence of per-window embedding vectors from
571 each session, separately. For each given recall trajectory, we then computed the cosine similarity
572 between the embedding vector for each recall window and the event-average embedding vector
573 for each of the 20 episode events. This yielded a number-of-windows by number-of-episode-
574 events (20) similarity matrix. We then treated the episode events as the hidden states of a hidden
575 Markov model with fixed parameters, and treated each recall window’s profile of similarities
576 to the episode events, z-scored across all window-event pairs, as its distribution of emission
577 scores over states. We imposed an ergodic transition structure (i.e., we allowed transitions from
578 any state to any other state, since participants may recall the episode events out of order, and
579 allowed states to be re-entered multiple times, since participants may revisit earlier events in
580 their recalls) governed by separate additive transition penalties for forward-one transitions and
581 all other transitions. Specifically, we imposed no penalty for remaining in the current state, a
582 small penalty (1.5 normalized emissions units) for transitioning from a given state to the next
583 (i.e., transitioning from recalling episode event n to recalling episode event $n + 1$), and a larger
584 penalty (3.0 normalized emissions units) for all other transitions. Finally, we decoded the most

585 probable sequence of episode-event labels using the Viterbi algorithm (Viterbi, 1967), collapsed
586 runs of consecutive identically labeled recall windows into events, and averaged the embedding
587 vectors over windows within each event. This procedure yielded, for each participant’s recall from
588 each session, a sequence of event-average embedding vectors and a set of labels denoting which
589 episode event each recall event corresponded to.

590 **Comparing trajectory structures**

591 We compared the geometric structures of the episode and recall event-sequences using dynamic
592 time warping (DTW; Sakoe and Chiba, 1978; Vintsyuk, 1968). The DTW algorithm finds the optimal
593 order-preserving alignment between two sequences of vectors with potentially different lengths or
594 sampling rates by locally “stretching” and “compressing” them in time to maximize their overall
595 similarity. This alignment yields a “warp path” that maps each vector in each sequence onto
596 one or more consecutive vectors in the other sequence with the minimum total distance between
597 matched vectors. The summed or average distance between matched vectors along the warp path
598 (the “warp cost”) is commonly taken as a measure of the two sequences’ structural dissimilarity.
599 We computed the warp cost as the average cosine distance between DTW-matched events so that
600 the cost values would be comparable across comparisons involving event-sequences of different
601 lengths (whereas summed distance necessarily grows with sequence length).

602 **Data availability**

603 All of the data analyzed in this manuscript may be found at [https://github.com/ContextLab/memory-](https://github.com/ContextLab/memory-dynamics)
604 [dynamics](https://github.com/ContextLab/memory-dynamics).

605 **Code availability**

606 All of the code for running our experiment and carrying out the analyses may be found at
607 <https://github.com/ContextLab/memory-dynamics>.

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